higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

LOGIC SYSTEMS N4

7 AUGUST 2019

This marking guideline consists of 9 pages.

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QUESTION 1

1.1 0 , 1 , 2 , 3 , 4 , 5, 6 , 7 , 8 , 9 , A , B , C , D , E , F (1)

1.2 Nonweighted positional code (2)

1.3 Excessive 3 code. (1)

1.4
- In asynchronous counters, the output of the previous flip-flop clocks the next one but synchronous counters are clocked simultaneously.
- Synchronous counters are faster than asynchronous ones. (2)

1.5 1.5.1 $A\overline{B}+\overline{A}B=A\oplus B$ ✓

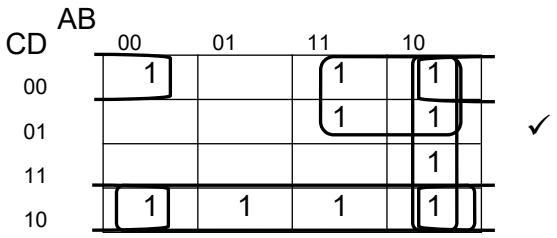
1.5.2 $A(AB+C)=AB+AC$ ✓

1.5.3 $A\overline{A}+AA=A$ ✓

1.5.4 $AB+B\overline{C}+\overline{A}+1=1$ ✓

(4 × 1) (4)

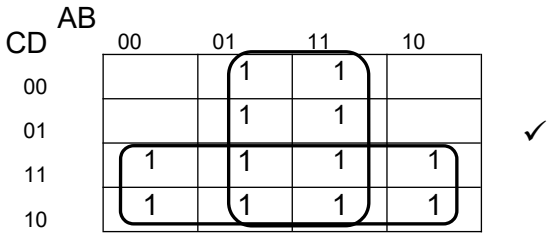
1.6 1.6.1



✓

$F=A\overline{C}+A\overline{B}+C\overline{D}+\overline{B}\overline{D}$ ✓

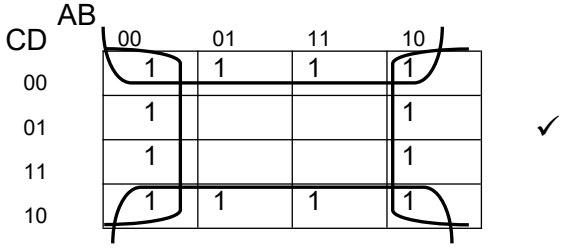
1.6.2



✓

$F=B+C$ ✓

1.6.3

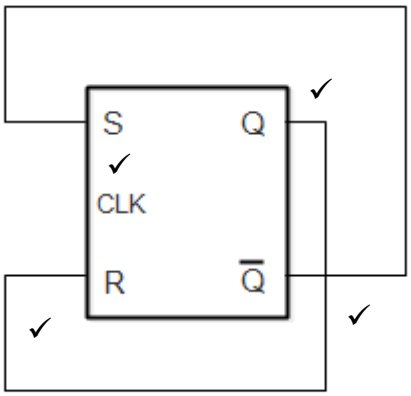


✓

$F=\overline{B}+\overline{D}$ ✓

(3 × 2) (6)

1.7



(4)
[20]

QUESTION 2

2.1 2.1.1 $34,6_8 - 38,875_{10}$

$34,6_8 = 011100,110_2$ ✓½

$38,875_{10} = 100110,111_2$ ✓½

$$\begin{array}{r} 011100,110 \\ -100110,111 \\ \hline \end{array}$$

$$\begin{array}{rr} & 011001,000 \checkmark \\ 0c & +011100,110 \\ \hline & 110101,110 \checkmark \\ 1's: & -001010,001 \checkmark \end{array}$$

$1010,001_2 = -\checkmark A,2_{16}\checkmark$

2.1.2 $12,1_8 \times 10,3_{10}$

$12,1_8 = 001010,001_2$ ✓½

$10,3_{10} = 1010,010_2$ ✓½

$$\begin{array}{rr} & 1010,010 \\ \times & 1010,001 \\ \hline & 1010010 \checkmark \\ & 10100100000 \checkmark \\ + & 1010010000000 \checkmark \\ \hline & 1100111,110010 \checkmark \end{array}$$

$1100111,110010 = 147,62_8$ ✓

2.1.3 $146,3_8 \div 23,C_{16}$

$146,3_8 = 1100110,011_2$ ✓½

$23,C_{16} = 10001,11_2$ ✓½

$$\begin{array}{r} 10,1 \checkmark \\ 10001111\overline{)110011001,1} \\ \underline{-10001111} \checkmark \\ 100000111 \\ \underline{-10001111} \checkmark \\ 1111000 \checkmark \end{array}$$

$10,1 = 2,5_{10}$ ✓

(3 × 6) (18)

2.2 2.2.1 1001 0111 1010

= 7✓ 1✓ 6✓ (3)

2.2.2 168_{10}

$= 10101000_2$ ✓

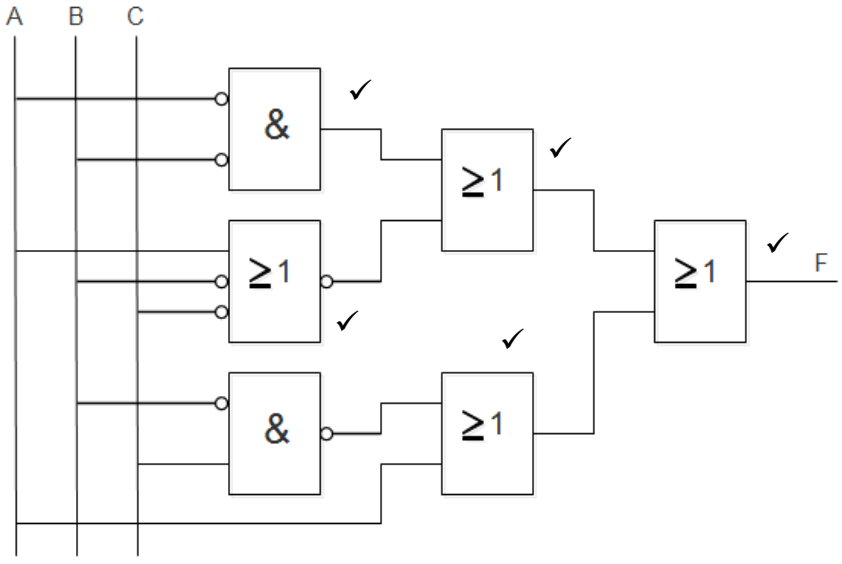
1+ 0+1+0+1+0+ 0+ 0 Binary✓

1 1 1 1 1 1 0 0 Gray✓✓ (4)

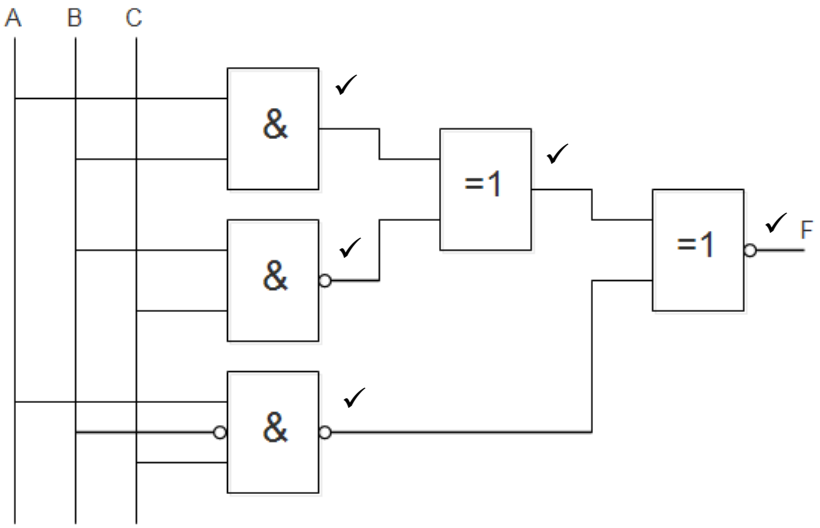
[25]

QUESTION 3

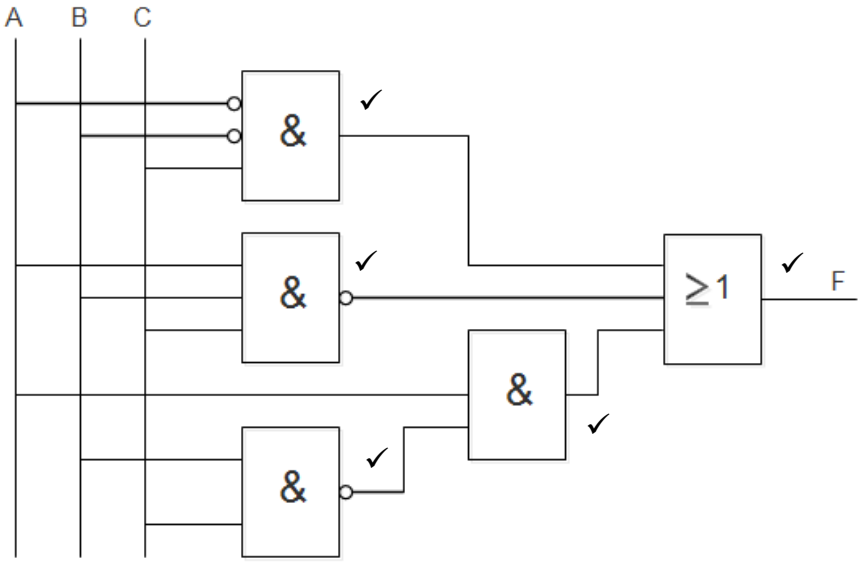
3.1 3.1.1



3.1.2



3.1.3

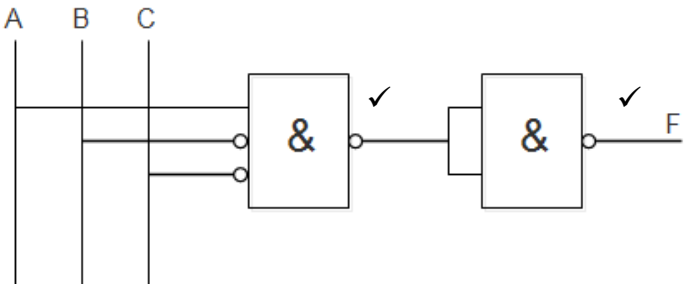


(3 × 5) (15)

3.2 3.2.1

$$F = A(\bar{A}+\bar{B}+C)(\bar{A}+B+\bar{C})(\bar{A}+\bar{B}+\bar{C})$$
$$= (A\bar{B}+AC)(\bar{A}+B+\bar{C})(\bar{A}+\bar{B}+\bar{C}) \quad \checkmark^{1/2}$$
$$= (A\bar{B}\bar{C}+ABC)(\bar{A}+\bar{B}+\bar{C}) \quad \checkmark^{1/2}$$
$$= A\bar{B}\bar{C}+A\bar{B}\bar{C} \quad \checkmark^{1/2}$$
$$= A\bar{B}\bar{C} \quad \checkmark^{1/2}$$

$$= \overline{\overline{A\bar{B}\bar{C}}} \quad \checkmark$$

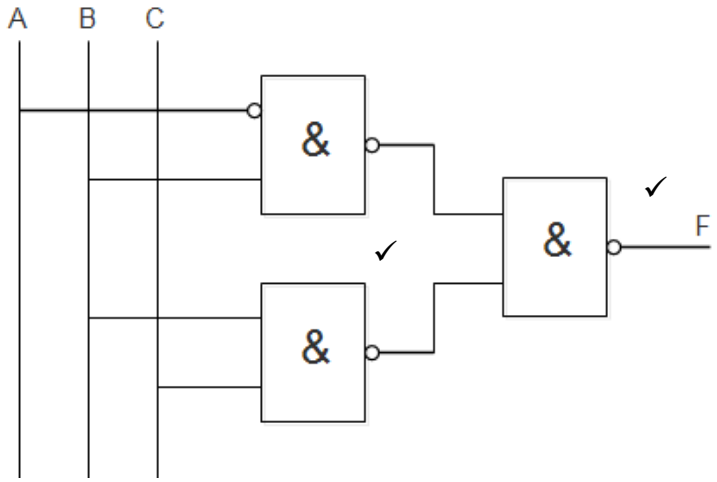


✓ ✓

3.2.2

$$F = \overline{\overline{(\bar{A}B+ABC)(\bar{A}BC+ABC)}}\ \overline{\overline{(\bar{A}B+ABC)}}$$
$$= (\bar{A}B+ABC)(\bar{A}BC+ABC)+(\bar{A}B+ABC) \quad \checkmark^{1/2}$$
$$= \bar{A}BC+ABC+\bar{A}B+ABC$$
$$= \bar{A}B(C+1)+ABC \quad \checkmark^{1/2}$$
$$= \bar{A}B+ABC$$
$$= B(\bar{A}+AC)$$
$$= B(\bar{A}+C)$$
$$= \bar{A}B+BC \quad \checkmark^{1/2}$$

$$= \overline{\overline{\bar{A}B+BC}} \quad \checkmark^{1/2}$$
$$= \overline{\overline{\bar{A}B}.\overline{BC}} \quad \checkmark$$

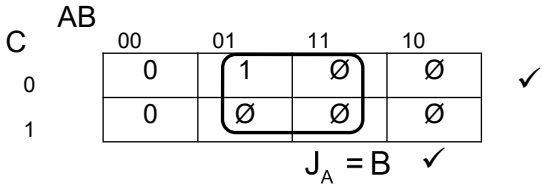


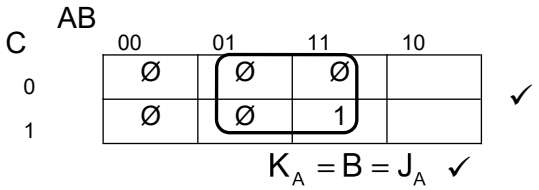
(2 × 5) (10)
[25]

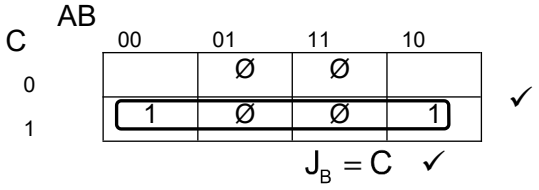
QUESTION 4

A	B	C	J_A	K_A	J_B	K_B	J_C	K_C
0	0	0	0	Ø	0	Ø	1	Ø
0	0	1	0	Ø	1	Ø	Ø	1
0	1	0	1	Ø	Ø	1	0	Ø
1	0	0	Ø	0	0	Ø	1	Ø
1	0	1	Ø	0	1	Ø	Ø	0
1	1	1	Ø	1	Ø	1	Ø	1
			✓		✓		✓	

011 - Ø 110 - Ø







C \ AB	00	01	11	10
0	Ø	1	Ø	Ø
1	Ø	Ø	1	Ø

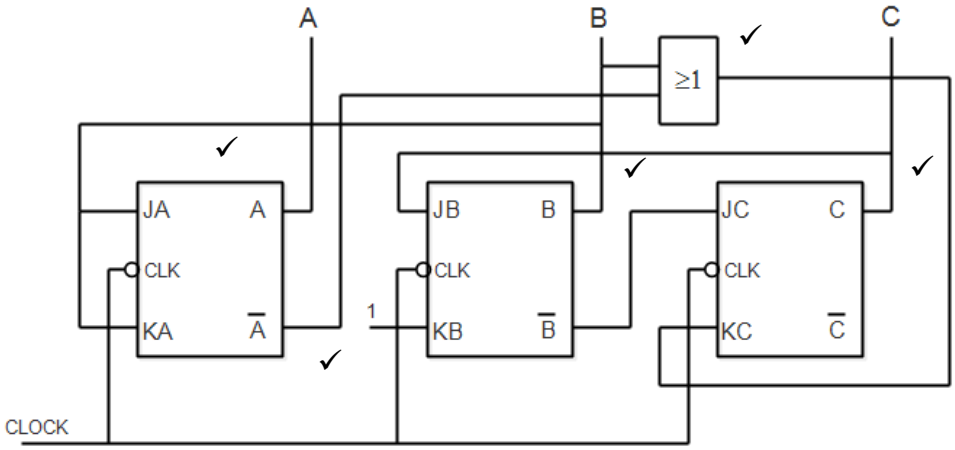
$K_B = 1$ ✓

C \ AB	00	01	11	10
0	1		Ø	1
1	Ø	Ø	Ø	Ø

$J_C = \overline{B}$

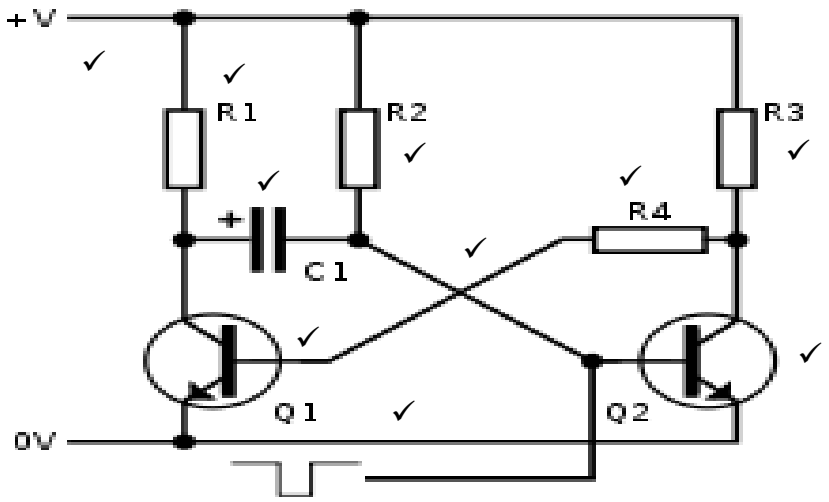
C \ AB	00	01	11	10
0	Ø	Ø	Ø	Ø
1	1	Ø	1	

$K_C = \overline{A} + B$ ✓



[20]

QUESTION 5



[10]

TOTAL: 100